

Diffusion and Flow-Based Models for Controllable Image Generation: A Survey

Diffusion and flow-based models have become dominant families for high-fidelity controllable image generation. They support text-to-image synthesis, structural conditioning, image editing, personalization, inpainting, and rapid sampling through a growing collection of guidance, adapter, flow matching, rectified-flow, and distillation methods. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around diffusion and score-based foundations, flow and continuous-time generative models, text conditioning and guidance, structural controls, editing and personalization, parameter-efficient adaptation, sampling acceleration, evaluation, and safety. The survey argues that controllability is not a single mechanism but a stack of conditioning signals, model interfaces, optimization objectives, and verification protocols. We close with open problems in faithful prompt following, compositional control, edit locality, data provenance, safety filtering, and reproducible evaluation.

Keywords:

diffusion models; flow matching; controllable image generation; text-to-image synthesis; ControlNet; image editing; guidance

1 Introduction

Image generation has shifted from adversarial and autoregressive systems toward diffusion, score-based, and continuous-time flow formulations. These models generate images by transforming noise into samples through learned denoising, score estimation, velocity fields, or probability flows. Their practical importance comes not only from fidelity but from controllability: users can steer images with text, layouts, masks, poses, sketches, depth maps, reference images, style examples, and editing instructions [24, 31, 40, 88, 148, 199, 256, 264]. Controllable generation is therefore a problem of aligning a generative trajectory with semantic and structural constraints.

Diffusion models provide an especially flexible control interface. Guidance can bias samples toward text prompts or class labels; cross-attention can localize prompt influence; adapters can inject spatial controls; inversion and editing methods can modify existing images; and personalization can bind a specific subject or style to a small set of examples [24, 31, 40, 88, 148, 199, 256, 264]. Flow matching and

rectified-flow models offer an alternative continuous-time view in which generation is learned as a transport process, often with attractive sampling and training properties [24, 31, 40, 88, 148, 199, 256, 264].

This survey reviews diffusion and flow-based models for controllable image generation. We focus on mechanisms that make generation steerable, inspectable, and reliable: conditioning, guidance, structural control, editing, personalization, adaptation, sampling, evaluation, and safety. The central claim is that controllability is a systems property spanning data, model architecture, conditioning interface, sampler, and human evaluation.

2 Background

2.1 Diffusion and Score-Based Models

Diffusion models define a forward corruption process and learn a reverse denoising process. Score-based formulations learn gradients of the data density and connect generation to stochastic differential equations and probability-flow ODEs. La-

tent diffusion compresses the image space before denoising, making high-resolution generation more tractable. These foundations enable strong image priors while allowing conditioning signals to be injected at several points in the generation trajectory [18, 24, 31, 36, 40, 148, 184, 264].

2.2 Flow-Based and Continuous-Time Generative Models

Flow-based generative models learn invertible or continuous transformations between simple noise and data. Flow matching and rectified flows train vector fields that transport samples along continuous paths, offering a bridge between normalizing flows, diffusion, and ODE-based sampling. Their relevance to controllable image generation lies in faster sampling, stable training objectives, and alternative ways to condition trajectories [24, 31, 40, 88, 148, 199, 256, 264].

2.3 What Control Means in Image Generation

Control can mean semantic control over object identity, stylistic control over appearance, spatial control over layout, local control over edits, or procedural control over the sampler. A method may succeed on one axis while failing another: it may follow a prompt but ignore layout, preserve identity but change background, or satisfy a pose map while reducing photorealism. Evaluation must therefore specify the control target [40, 109, 132, 137, 167, 184, 190, 282].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with diffusion models, controllable image generation, text-to-image diffusion, latent diffusion, score-based generative models, classifier-free guidance, guided diffusion, ControlNet, T2I-Adapter, pose/depth/layout/segmentation/sketch conditioning, image editing, inpainting, personalization, DreamBooth, textual inversion, LoRA, flow matching, rectified flows, normalizing flows, consistency models, diffusion distillation, one-step and few-step sampling, evaluation, safety, watermarking, and stable diffusion control terms. Records were retained when title or abstract evi-

dence connected diffusion, score-based, flow-based, or continuous-time generative models with image generation, conditioning, control, editing, sampling, evaluation, or safety. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the field along six axes. The first is *generative dynamics*: discrete diffusion, score-based SDE, latent diffusion, probability-flow ODE, flow matching, rectified flow, normalizing flow, or consistency model. The second is *conditioning signal*: text, class, mask, edge, depth, pose, segmentation, layout, reference image, style, identity, or reward model. The third is *control locus*: training objective, model architecture, cross-attention, adapter, sampler guidance, inversion, or post-hoc optimization. The fourth is *task*: text-to-image generation, image editing, inpainting, personalization, style transfer, super-resolution, or domain adaptation. The fifth is *adaptation*: full fine-tuning, LoRA, textual embeddings, adapters, distillation, or preference alignment. The sixth is *evaluation*: fidelity, prompt alignment, spatial accuracy, identity preservation, edit locality, diversity, speed, safety, and provenance.

3.3 Text Conditioning and Guidance

Text-to-image systems condition generation on prompts through text encoders, cross-attention, classifier guidance, classifier-free guidance, or learned preference models. Guidance improves prompt alignment but can reduce diversity or introduce artifacts when pushed too strongly. Prompt control is also compositional: the model must bind attributes to objects, count instances, preserve relations, and avoid prompt leakage into unrelated regions [24, 31, 40, 88, 148, 199, 256, 264].

3.4 Structural Controls

Structural controls provide spatial information beyond text. Edges, depth maps, human poses, segmentation maps, bounding boxes, layouts, sketches, and masks can constrain composition. Control modules such as adapters and ControlNet-style branches preserve pretrained image priors while injecting additional conditions. The hard part is balancing control strength with visual plausibility: strict conditioning can overfit the guide image, while weak conditioning ignores user intent [24, 36, 40, 88, 148, 199, 209, 256].

3.5 Image Editing, Inpainting, and Personalization

Editing methods modify an existing image while preserving irrelevant content. Inpainting fills masked regions; prompt-based editing changes semantics; inversion maps an image into the model’s latent or noise trajectory; personalization binds a subject, identity, or style to a new prompt. These tasks expose the locality problem: a good edit should change what the user requested without drifting unrelated objects, lighting, or identity [24, 123, 148, 166, 184, 199, 256, 257].

3.6 Training and Parameter-Efficient Adaptation

Controllable generation often requires adapting a large pretrained model. Full fine-tuning is expensive and can degrade general capability; parameter-efficient methods such as LoRA, adapters, learned embeddings, and lightweight control branches reduce cost. Distillation and preference optimization further adapt models for speed or human judgments. The open question is how to add control without memorizing small datasets or weakening diversity [24, 31, 40, 88, 148, 199, 256, 264].

3.7 Sampling, Acceleration, and Consistency

Diffusion sampling can be slow because many denoising steps are required. Solvers, distillation, consistency models, rectified flows, and one-step or few-step samplers reduce latency. Faster sampling changes controllability: fewer steps leave less room for guidance, correction, or iterative editing. Future samplers should therefore be evaluated on control accuracy, not only image quality and speed [18, 21, 36, 123, 148, 166, 167, 199].

3.8 Evaluation and Safety

Evaluation includes image fidelity, diversity, prompt following, spatial control accuracy, identity preservation, edit locality, human preference, robustness, and inference cost. Safety and provenance are also central because generative models can reproduce harmful stereotypes, create deceptive images, leak training data, or obscure copyright and consent. Watermarking, dataset documentation, filtering, and red-team evaluation are part of the controllability stack [40, 109, 132, 137, 167, 184, 190, 282].

3.9 Representative Literature Map

The following map cites the full retained corpus by theme, making the survey coverage explicit.

Diffusion and score-based generative foundations

The diffusion and score-based generative foundations cluster contains work that develops methods, systems, datasets, or analyses central to this survey [6, 18, 20, 21, 24, 36, 142, 160, 184]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [1, 22, 63, 65, 84, 109, 124, 165, 234]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [57, 62, 76, 105, 135, 141, 179, 222, 243]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [34, 126, 133, 136, 183, 224, 265, 266, 270]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [58, 75, 125, 145, 164, 178, 188, 217, 269]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [48, 71, 94, 97, 158, 170, 196, 241, 300]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [78, 116, 129, 189, 197, 229, 237, 263, 284]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [51, 100, 111, 156, 180, 192, 258, 261, 281]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [28, 64, 73, 80, 83, 144, 193, 231, 296]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [45, 53, 60, 66, 72, 161, 162, 185, 255]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [5, 74, 104, 151, 205, 206, 216, 227, 240]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [46, 61, 92, 103, 114, 120, 139, 154, 295]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [19, 35, 101, 251].

Flow matching, rectified flows, and continuous-time models

The flow matching, rectified flows, and continuous-time models cluster contains work that

develops methods, systems, datasets, or analyses central to this survey [167, 245].

Text conditioning, guidance, and prompt control The text conditioning, guidance, and prompt control cluster contains work that develops methods, systems, datasets, or analyses central to this survey [40, 88, 166, 190, 212, 239, 256, 257, 264]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [55, 67, 107, 112, 127, 171, 213, 273, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [7, 25, 47, 68, 96, 169, 236, 262, 291]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 9, 17, 37, 50, 82, 90, 203, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 41, 81, 106, 110, 210, 218, 223, 233]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [2, 69, 177, 187, 198, 250, 272, 286, 288]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [30, 117, 143].

Structural controls and conditional generation The structural controls and conditional generation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [56, 59, 70, 123, 132, 148, 182, 201, 209]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [15, 23, 42, 118, 147, 252, 280, 292, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [33, 130, 134, 191, 225, 246, 253, 283, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [11, 39, 43, 99, 119, 159, 176, 221, 275]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [49, 52, 108, 115, 163, 211, 226, 228, 244]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [79, 93, 153, 208, 219, 230, 238, 287, 294]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and de-

ployment assumptions [4, 14, 85, 95, 122, 149, 172, 254, 271]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [29].

Image editing, inpainting, and personalization The image editing, inpainting, and personalization cluster contains work that develops methods, systems, datasets, or analyses central to this survey [13, 16, 26, 91, 128, 199, 214, 268, 276].

Training, fine-tuning, and parameter-efficient adaptation The training, fine-tuning, and parameter-efficient adaptation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [32, 86, 113, 121, 155, 174, 204, 232, 242]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [10, 77, 87, 131, 168, 186, 195, 278, 279]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [267].

Sampling, acceleration, and consistency models The sampling, acceleration, and consistency models cluster contains work that develops methods, systems, datasets, or analyses central to this survey [27, 44, 54, 157, 235, 249, 298].

Benchmarks, evaluation, safety, and provenance The benchmarks, evaluation, safety, and provenance cluster contains work that develops methods, systems, datasets, or analyses central to this survey [31, 137, 138, 152, 173, 181, 202, 259, 282]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [8, 38, 175, 194, 220, 247, 260, 277, 297]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [89, 98, 140, 146, 150, 207, 215, 274, 299]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [102, 200, 248].

4 Challenges and Prospects

4.1 Compositional Prompt Following

Models still struggle with binding attributes, counting objects, preserving relations, and satisfying multiple controls at once. Progress will require better training data, stronger grounding objectives, and

evaluations that test compositional control rather than single-object prompts.

4.2 Control Strength, Locality, and Tradeoffs

Every control mechanism has a strength knob, explicit or implicit. Too little control ignores the user; too much control reduces realism or diversity. Editing adds locality: modifications should remain confined to intended regions. Future interfaces should expose these tradeoffs rather than hiding them inside sampler settings.

4.3 Flow Models and Fast Controllable Sampling

Flow matching, rectified flows, and consistency-style training may reduce sampling cost, but the field still needs to understand how fast trajectories interact with guidance and structural controls. Fast models should be compared on controllability, editability, and safety, not only throughput.

4.4 Safety, Provenance, and Dataset Governance

Controllable image generation raises questions about deception, consent, copyright, bias, and watermark robustness. Provenance-aware generation should record model, prompt, conditioning inputs, edits, and sampling settings so that images can be audited after creation.

5 Conclusion

Diffusion and flow-based models have made controllable image generation a mature but still rapidly changing field. This survey reviewed diffusion and score-based foundations, flow matching, text guidance, structural conditioning, editing, personalization, adaptation, fast sampling, evaluation, and safety. The central lesson is that control is distributed across the entire generation pipeline. Reliable systems must coordinate conditioning interfaces, model architectures, samplers, evaluation protocols, and governance mechanisms.

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